

OUTLIER SYNTHESIS VIA HAMILTONIAN MONTE CARLO FOR OUT-OF-DISTRIBUTION DETECTION

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ABSTRACT

Out-of-distribution (OOD) detection is crucial for developing trustworthy and reliable machine learning systems. Recent advances in training with auxiliary OOD data demonstrate efficacy in enhancing detection capabilities. Nonetheless, these methods heavily rely on acquiring a large pool of high-quality natural outliers. Some prior methods try to alleviate this problem by synthesizing virtual outliers but suffer from either poor quality or high cost due to the monotonous sampling strategy and the heavy-parameterized generative models. In this paper, we overcome all these problems by proposing the *Hamiltonian Monte Carlo Outlier Synthesis* (HamOS) framework, which views the synthesis process as sampling from Markov chains. Based solely on the in-distribution data, the Markov chains can extensively traverse the feature space and generate diverse and representative outliers, hence exposing the model to miscellaneous potential OOD scenarios. The Hamiltonian Monte Carlo with sampling acceptance rate almost close to 1 also makes our framework enjoy great efficiency. By empirically competing with SOTA baselines on both standard and large-scale benchmarks, we verify the efficacy and efficiency of our proposed HamOS. Our code is available at: https://github.com/Fir-lat/HamOS_OOD.

1 INTRODUCTION

Despite the impressive achievements of deep neural networks across various practical applications, they are unable to make dependable predictions when faced with *out-of-distribution* (OOD) samples (Nguyen et al., 2015; Bendale & Boulton, 2016; Hendrycks & Gimpel, 2017), as the input data may not be drawn from *in-distribution* (ID) (Bommasani et al., 2021). To guarantee the reliability of machine learning systems, models should recognize data from unknown classes instead of making overconfident predictions, known as OOD detection. It has garnered significant attention within the Safety AI community, as erroneous predictions of OOD samples might result in perilous situations, especially vital for safety-critical applications like autonomous driving (Heidecker et al., 2021a;b) and medical image analysis (Ulmer et al., 2020; Linmans et al., 2020).

Prior research can be categorized into two paradigms: post-hoc methods and regularization-based methods. The former focuses on designing a robust scoring function to differentiate between ID and OOD samples (Hendrycks & Gimpel, 2017; Liang et al., 2018; Liu et al., 2020; Mahalanobis, 1936; Sun et al., 2022), while the latter incorporates supplementary regularization during the training phase, with or without auxiliary OOD data, to enhance the model’s discriminative ability (Hendrycks et al., 2019; Wei et al., 2022; Zhang et al., 2023a; Zhu et al., 2023b; Ming et al., 2023a), typically achieving better performance than post-hoc methods.

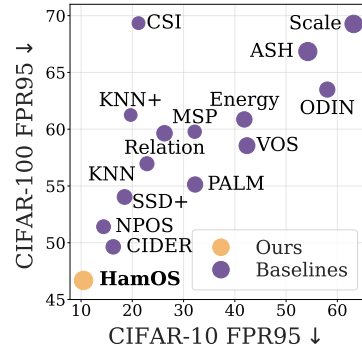


Figure 1: OOD detection performance on CIFAR and ImageNet. The size of the dots indicates AU-ROC w.r.t. ImageNet-1K.

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